

Lecture 4: Some Principles of Large Sample Theory

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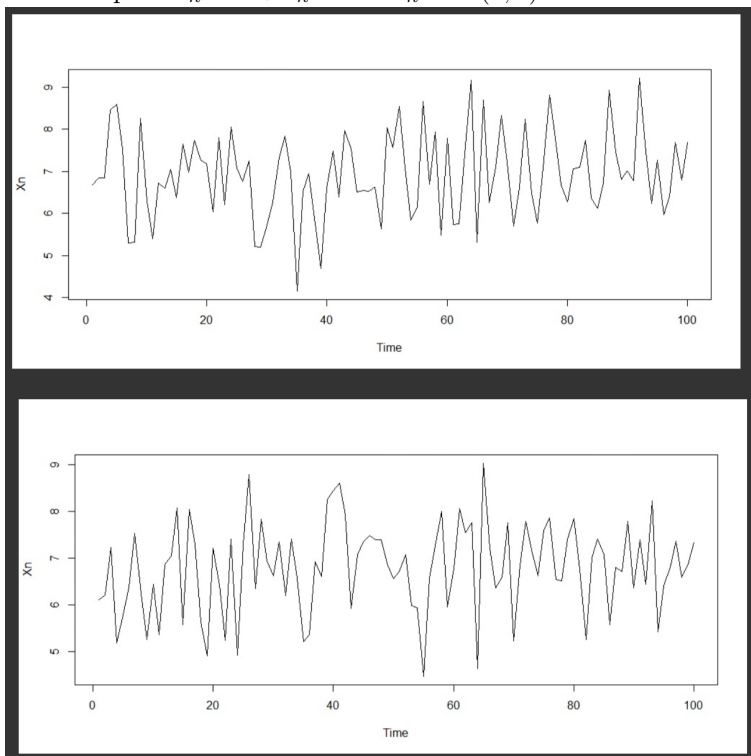
1 Random Sequences & Random Convergence

Definition 4.1 Let (Ω, β, P) be a probability space. A sequence of random vectors is a infinite countable set of random vectors defined on the same probability space, i.e. $\{X_n\}_{n=1}^{\infty}$ where for $n \in N$, $X_n : \Omega \rightarrow R^k$

$[R^k$ denotes k -dimensional Euclidean space: $x_1, x_2, \dots, x_k]$

If $\omega \in \Omega$ represents a specific outcome, then $\{X_n(\omega)\}_{n=1}^{\infty}$ represents a specific realization of the sequence, i.e., it consists of a specific sequence of real vectors. $\{X_n(\omega)\}_{n=1}^{\infty}$ is called a sample path of the random sequence.

Example: $X_n = 7 + \varepsilon_n$ where $\varepsilon_n \sim N(0, 1)$



Both of these pictures represent the first 100-terms of realization of the random sequence $\{X_n\}$. Note that both realizations are different and they don't tend to any value.

Consider now the sequence of averages:

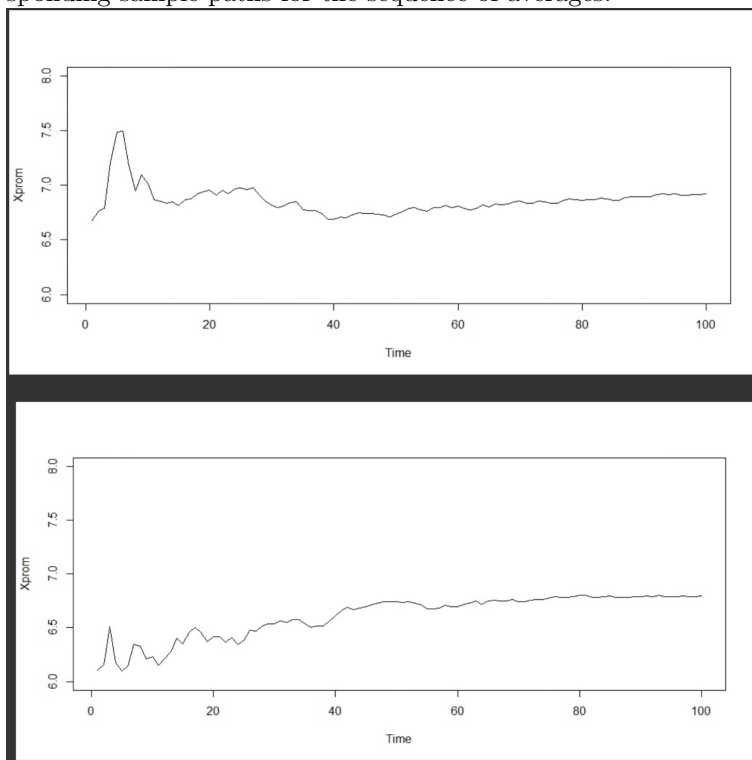
$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

$$\bar{X}_1 = X_1$$

$$\bar{X}_2 = \frac{1}{2}(X_1 + X_2)$$

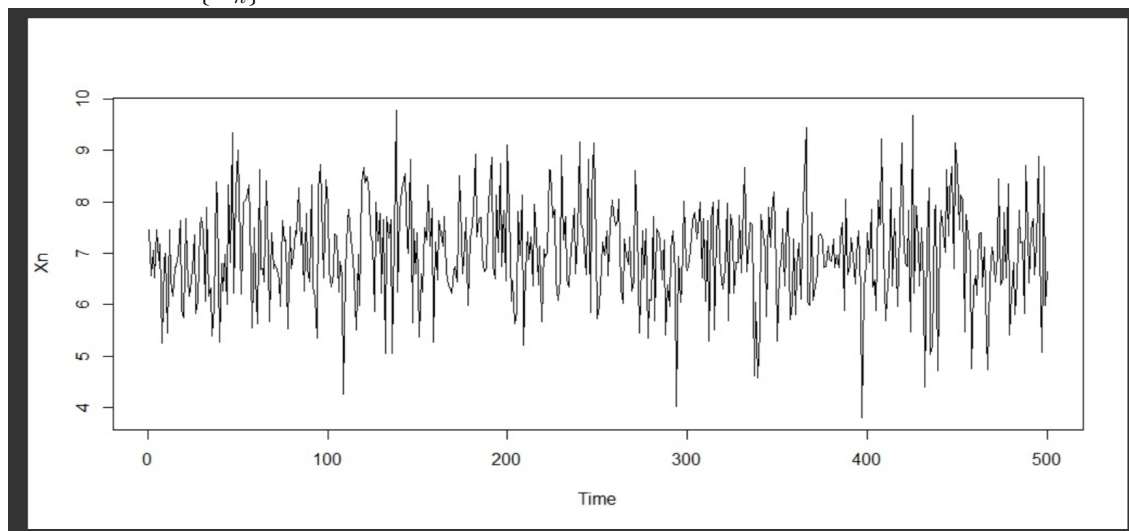
$$\bar{X}_3 = \frac{1}{3}(X_1 + X_2 + X_3)$$

For the sample paths in the previous slide, the figures here represent the corresponding sample paths for the sequence of averages.

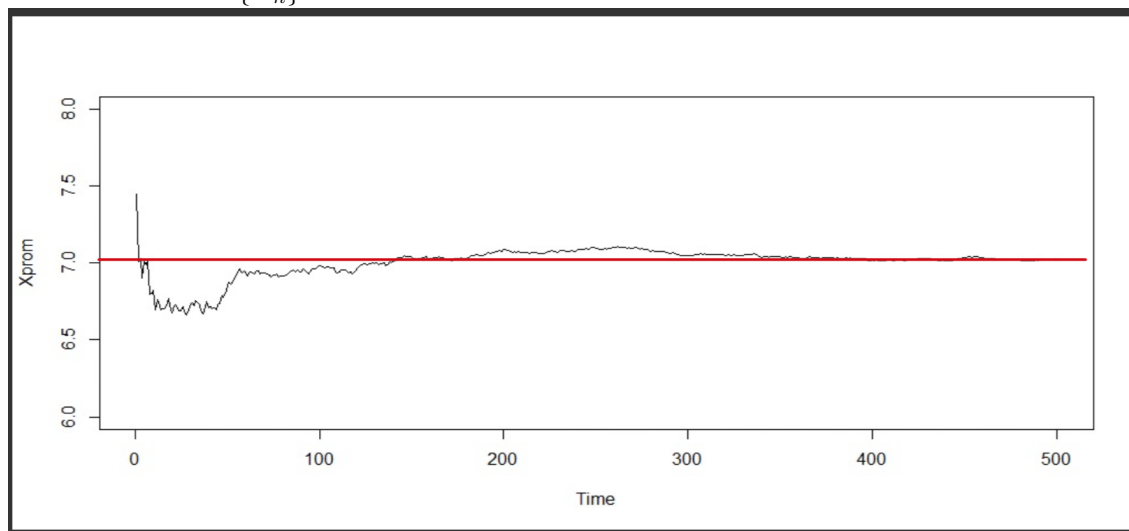


In here, both realisations seem to approach the number 7.

Same example, but first 500 terms.
A realization of $\{X_n\}$



A realization of $\{\bar{X}_n\}$



In the case of a sequence of real numbers $\{a_n\}$, there is a precise definition of what is meant by convergence to a number a .

Definition: The sequence a_n converges to the number a if for every $\varepsilon > 0$ there exists an index $N(\varepsilon)$ such that $|a_n - a| < \varepsilon$ for any $n \geq N(\varepsilon)$

No matter how narrow is a neighborhood of radius $\varepsilon > 0$ around a (the interval $(a - \varepsilon, a + \varepsilon)$) there is always an index $N(\varepsilon)$ such that the whole tail of the sequence is inside the neighborhood.

→ We will come up with a probabilistic definition of convergence.

Notation: $|X_n - c|$ stands the absolute value if $\{X_n\}$ is a sequence of random variables or is a vector norm for a sequence of random vectors, and in this case c is a vector too.

Definition 4.2 It is said that a random sequence $\{X_n\}$ converges in probability to a constant c if for every $\varepsilon > 0$

$$P(|X_n - c| > \varepsilon) \longrightarrow 0 \text{ when } n \rightarrow \infty$$

OR equivalently

$$P(|X_n - c| \leq \varepsilon) \longrightarrow 1 \text{ when } n \rightarrow \infty$$

This convergence is denoted by

$$X_n \xrightarrow{P} c$$

Convergence in Probability is hard to apply discretely since $P(|X_n - c| > \varepsilon)$ depends on the pdf of X_n .

In order to avoid such computation another type of convergence is also considered, this type of convergence depends on the second moment of X_n .

Definition 4.3 It is said that a random sequence $\{X_n\}$ converges in quadratic mean to a constant c if

$$E\{|X_n - c|^2\} \rightarrow 0 \text{ when } n \rightarrow \infty$$

This convergence is denoted by

$$X_n \xrightarrow{L_2} c$$

It is important to link convergence in quadratic mean and convergence in probability.

Chebychev's Inequality. Let X be a random variable and let $g(x)$ be a non-negative function. Then, for any $r > 0$

$$P(g(X) \geq r) \leq \frac{E[g(X)]}{r}$$

Proof:

$$\begin{aligned} E[g(X)] &= \int_{-\infty}^{\infty} g(x) f_x(x) dx \\ &\geq \int_{\{x: g(x) \geq r\}} g(x) f_x(x) dx \\ &\geq \int_{\{x: g(x) \geq r\}} r f_x(x) dx \\ &= r P(g(X) \geq r) \end{aligned}$$

$$\begin{aligned} \Rightarrow E[g(X)] &\geq r P(g(X) \geq r) \\ \Rightarrow \frac{E[g(X)]}{r} &\geq P(g(X) \geq r) \end{aligned}$$

■

Markov's Inequality. If X is a non-negative random variable and $a > 0$, then

$$P(X \geq a) \leq \frac{E[X]}{a}$$

Proof: This is just a restatement of Chebychev's Inequality.

Theorem 4.1. If $X_n \xrightarrow{L_2} c$ then $X_n \xrightarrow{P} c$

Proof: Clearly the random variable $|X_n - c|^2$ is non-negative and its expectation exists from some n onwards, then from Markov's inequality:

$$P(|X_n - c| \geq \varepsilon) = P(|X_n - c|^2 \geq \varepsilon^2) \leq \frac{E[|X_n - c|^2]}{\varepsilon^2} \rightarrow 0 \text{ when } n \rightarrow \infty$$

$\therefore$ The result is true.

■

Corollary 4.2. Let $\{T_n\}$ be a sequence such that $\lim_{n \rightarrow \infty} E[T_n] = c$ and $\lim_{n \rightarrow \infty} Var[T_n] = 0$. Then:

$$T_n \xrightarrow{P} c$$

Proof: Homework!

Theorem 4.3. Let $\{T_n\}$ be a sequence such that $T_n \xrightarrow{P} c$ and g is a continuous function at c , then:

$$g(T_n) \xrightarrow{P} g(c)$$

Proof: Given $\varepsilon > 0$ we want to show that

$$P(|g(T_n) - g(c)| > \varepsilon) \rightarrow 0 \text{ when } n \rightarrow \infty$$

OR equivalently

$$P(|g(T_n) - g(c)| \leq \varepsilon) \rightarrow 1 \text{ when } n \rightarrow \infty$$

From continuity of g at c , there exists a $\sigma > 0$ that depends on ε and such that $|g(t) - g(c)| < \varepsilon$ whenever $|t - c| < \sigma$. The previous implies that:

$$\{|T_n - c| < \sigma\} \subseteq \{|g(T_n) - g(c)| < \varepsilon\}$$

Taking Probabilities:

$$P(|T_n - c| < \sigma) \leq P(|g(T_n) - g(c)| < \varepsilon) \leq 1 \quad (*)$$

but from hypothesis $P(|T_n| < \sigma) \rightarrow 1$ when $n \rightarrow \infty$

Therefore, taking limits in $(*)$, we have that

$$1 \leq P(|g(T_n) - g(c)| < \varepsilon) \leq 1 \text{ when } n \rightarrow \infty$$

$$\Rightarrow P(|g(T_n) - g(c)| < \varepsilon) \rightarrow 1 \text{ when } n \rightarrow \infty$$

Note that the sequences in Theorem 4.3 could be vector valued.
 The notion of convergence in probability is central to the notion of consistency of an estimator.

Definition 4.4. Let $\hat{\theta}_n$ be an estimator of the parameter θ , where n stands for the sample size. An estimator is **consistent** if

$$\hat{\theta}_n \xrightarrow{P} \theta$$

Remark: From Theorem 3.3 the LSE is a continuous function of some averages, so we need some results on the convergence in probability of sample averages

↓
 Weak Law of Large Numbers

2 Weak Law of Large Numbers

Weak vs Strong Law of Large Numbers: SLLN $\Rightarrow$ WLLN

Weak: Convergence in probability.

Strong: Almost sure convergence - stronger than convergence in probability.

$\rightarrow$ For our purposes in Econometrics, the WLLN is enough.

Theorem 4.4. Weak Law of Large Numbers (WLLN)

Let $\{X_i\}$ be such that $E(X_i) = \mu$ and $Var(X_i) = \sigma^2$ for any i . Also $Cov(X_i, X_j) = 0$ for $i \neq j$. Then

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{p} \mu$$

Proof:

$$\begin{aligned} E[\bar{X}_n] &= E\left(\frac{1}{n} \sum_i X_i\right) = \frac{1}{n} \sum E[X_i] = \mu \\ \text{Var}(\bar{X}_n) &= \text{Var}\left(\frac{1}{n} \sum_i X_i\right) = \frac{1}{n^2} \text{Var}\left(\sum_i X_i\right) \\ &= \frac{1}{n^2} \left(\sum_i \text{Var}(X_i) + \sum_{i \neq j} \text{Cov}(X_i, X_j) \right) \\ &= \frac{1}{n^2} (\sum_i \sigma^2 + 0) = \frac{\sigma^2}{n} \end{aligned}$$

Then

$$\begin{aligned} \lim_{n \rightarrow \infty} E[\bar{X}_n] &= \mu \text{ (trivial, because } E[\bar{X}_n] = \mu, \text{ i.e., a constant)} \\ \lim_{n \rightarrow \infty} \text{Var}(\bar{X}_n) &= \lim_{n \rightarrow \infty} \frac{\sigma^2}{n} = 0 \end{aligned}$$

$\therefore$ By Corollary 4.2, $\bar{X}_n \xrightarrow{P} \mu$

■

Example (continued):

$X_n = \mu + \varepsilon_n$, $\varepsilon \sim N(0, 1)$ for each $n \in \mathbb{N}$, $\mu = 7$

$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$ by WLLN, given this setup, $\bar{X}_n \rightarrow \mu = 7$

Example (A consistent estimate of σ^2)

Let $\{X_i\}$ be a sequence of i.i.d random variables with finite variance. We are going to show that a consistent estimator of $\sigma^2 = \text{Var}(X_1) = E(X_1^2) - \mu^2$ is

given by $\widehat{\text{Var}}(X) = \frac{1}{n} S_{xx} = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2 = \frac{1}{n} \sum_{i=1}^n X_i^2 - \bar{X}_n^2$

From Theorem 4.3 and Theorem 4.4 (WLLN), since $\bar{X}_n \xrightarrow{P} \mu$ and that $g(x) = x^2$

is continuous, then $(\bar{X}_n)^2 \xrightarrow{P} \mu^2$

Now, we need to see how to estimate $E(X_1^2)$
 Since X_i and X_j are independent, then also X_i^2 and X_j^2 are independent $\Rightarrow$
 $Cov(X_i^2, X_j^2) = 0$. Consider $Z_n = X_n^2$, then from Theorem 4.4 (WLLN)

$$\bar{Z}_n = \frac{1}{n} \sum_{i=1}^n Z_i = \frac{1}{n} \sum_{i=1}^n X_i^2 \xrightarrow{P} E(Z_i) = E(X_i^2)$$

$$\therefore (\bar{X}_n)^2 \xrightarrow{P} \mu^2, \bar{Z}_n \xrightarrow{P} E(X_i^2).$$

Consider the sequence of random vectors $(\bar{X}_n, \bar{Z}_n)$, $(\bar{X}_n, \bar{Z}_n) \xrightarrow{P} (\mu, E(X_1^2))$.

Consider the function $h(x, z) = z - x^2$ which is continuous everywhere. Then from Theorem 4.3

$$h(\bar{X}_n, \bar{Z}_n) = \bar{Z}_n - (\bar{X}_n)^2 \xrightarrow{P} g(\mu, E(X_1^2)) = E(X_1^2 - \mu^2) = \sigma^2$$

That is:

$$\widehat{Var(X)} = \frac{1}{n} S_{xx} = \frac{1}{n} \sum_{i=1}^n X_i^2 - (\bar{X}_n)^2 \xrightarrow{P} \sigma^2$$

$\therefore \widehat{Var(X)} = \frac{1}{n} S_{xx}$ is consistent.

Homework: Show that $\widehat{Cov(X, Y)} = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)(Y_i - \bar{Y}_n) = \frac{1}{n} S_{xy}$ converges in probability to $Cov(X, Y)$

We can show that $LSE(\hat{\beta}_0, \hat{\beta}_1)$ from simple linear regression are consistent estimators of the coefficients of the BLP under very mild assumptions.

Recall that the BLP $E^*(Y|X) = \beta_0 + \beta_1 X$, where $\beta_0 = E(Y) - \beta_1 E(X)$ and $\beta_1 = \frac{Cov(X, Y)}{Var(X)}$

Now consider the estimator for the slope $\hat{\beta}_1$

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{\frac{1}{n}S_{xy}}{\frac{1}{n}S_{xx}} = \frac{\widehat{Cov(X, Y)}}{\widehat{Var(X)}}$$

We know that (from previous results & homework):

$$\frac{1}{n}S_{xy} \xrightarrow{P} Cov(X, Y) \tag{1}$$

$$\frac{1}{n}S_{xx} \xrightarrow{P} Var(X) \tag{2}$$

Now $g(x, y) = \frac{y}{x}$ is continuous for $x > 0$, without loss of generality we can assume that $Var(X) > 0$.

Then since g is continuous for $x > 0$ by Theorem 4.3

$$g(S_{xx}/n, 1/n S_{xy}/n) = \frac{\frac{1}{n}S_{xy}}{\frac{1}{n}S_{xx}} \xrightarrow{P} g(Var(X), Cov(X, Y)) = \frac{Cov(X, Y)}{Var(X)} = \beta_1$$

because $(\frac{1}{n}S_{xx}, \frac{1}{n}S_{xy}) \xrightarrow{P} (Var(X), Cov(X, Y)) \therefore \hat{\beta}_1 \xrightarrow{P} \beta_1$

Homework: Show that $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$ converges in probability to $\beta_0 \therefore (\hat{\beta}_0, \hat{\beta}_1) \xrightarrow{P} (\beta_0, \beta_1) \Rightarrow (\hat{\beta}_0, \hat{\beta}_1)$ is a consistent estimator of (β_0, β_1) which are the parameters of the BLP. These results will be extended to multivariate BLP in Lecture 5.

3 Convergence in Distribution and the Central Limit Theorem

Definition 4.5. Let $\{X_n\}$ be a sequence of random variables with cumulative distribution functions given by $G_n(t) = P(X_n \leq t)$.

It is said that X_n converges in distribution to a sdf G if

$$\lim_{n \rightarrow \infty} G_n(t) = G(t)$$

for every t where G is continuous.

This mode of convergence is denoted by

$$X_n \xrightarrow{d} G, \text{ (G is cdf)}$$

Or it can also be denoted by

$$X_n \xrightarrow{d} X, \text{ where } X \text{ has cdf } G$$

(X is a r.v. with cdf G)

Remarks: In this type of convergence the values of the sequence do not necessarily converge to anything, instead it is the probability of the event that $X_n \leq t$ which is approximated by $G(t)$ when n is large. The CLT, intuitively, says: the standardized version of the sample average of a sequence of i.i.d. random variables converges in distribution to a standard normal distribution (or standard normal random variable).

Theorem 4.5 The (Lindeberg-Lévy) Central Limit Theorem (CLT)
 Let $\{X_n\}$ be a sequence of i.i.d. random variables with $E[X_n] = \mu$, $Var(X_n) = \sigma^2 < \infty$, let $Z_n = (\bar{X}_n - \mu)/\sqrt{\sigma^2/n}$ be the standardized average. Then $Z_n \xrightarrow{d} \Phi$, where Φ is the cdf of a $N(0,1)$

Theorem 4.5 can also be stated as $Z_n \xrightarrow{d} N(0, 1)$, or also as $Z_n \xrightarrow{d} Z$ where Z is distributed as $N(0, 1)$.

In practical terms, Theorem 4.5 produces the following approximation:

$$P(\bar{X}_n \leq x) = P(Z_n \leq \frac{x - \mu}{\sqrt{\sigma^2/n}}) \approx \Phi(\frac{x - \mu}{\sqrt{\sigma^2/n}}) \text{ for large } n$$

This approximation above can be described by saying that the average is approximately normal with mean μ and variance σ^2/n , and this can also be denoted by $\bar{X}_n \approx N(\mu, \sigma^2/n)$.

Note: $\approx$ is short-hand for "asymptotically normal distribution".

Similarly, we can obtain an asymptotic approximation for $\sum_{i=1}^n X_i$ since:

$$P\left(\sum_{i=1}^n X_i \leq a\right) = P\left(\frac{1}{n} \sum_{i=1}^n X_i \leq a/n\right) = P\left(\bar{X}_n \leq \frac{a}{n}\right)$$

then using the result from the previous slide:

$$P\left(\sum_{i=1}^n X_i \leq a\right) \approx \Phi\left(\frac{a/n - \mu}{\sqrt{\sigma^2/n}}\right) = \Phi\left(\frac{a - n\mu}{\sqrt{n\sigma^2}}\right)$$

$$\frac{\frac{a}{n} - \mu}{\frac{\sqrt{\sigma^2}}{\sqrt{n}}} = \frac{\frac{a - n\mu}{n}}{\frac{\sqrt{\sigma^2}}{\sqrt{n}}} = \frac{(a - n\mu)\sqrt{n}}{n\sqrt{\sigma^2}} = \frac{(a - n\mu)\sqrt{n}}{(\sqrt{n})^2\sqrt{\sigma^2}} = \frac{a - n\mu}{\sqrt{n\sigma^2}}$$

This can be interpreted by saying that $\sum_{i=1}^n X_i$ is approx. normal with mean $n\mu$ and variance $n\sigma^2$, and this can also be denoted $\sum_{i=1}^n X_i \approx N(n\mu, n\sigma^2)$

Example Suppose that the sequence of random variables X_i is i.i.d. with chi-square distribution with 1 degree of freedom. $X_i \sim \chi^2(1)$
 We want to compute the approximation provided by the CLT with the exact value of $P(\sum_{i=1}^{30} X_i \leq 34.8)$. We know $E[X_i] = 1 = \mu$, $Var(X_i) = 2 = \sigma^2$
 Then using the approximation we get:

$$P(\sum_{i=1}^{30} X_i \leq 34.8) \approx (\frac{34.8 - 30}{\sqrt{30 * 2}}) = \Phi(\frac{4.8}{\sqrt{60}}) = \Phi(0.62) = 0.73237$$

exact value since

$$\sum_{i=1}^n X_i \sim \chi^2(30) \text{ This is the exact distribution}$$

$$P(\sum_{i=1}^{30} X_i \leq 34.8) = P(\chi^2(30) \leq 34.8) = 0.75$$

Although n is not "large" but the approximation is good enough.

Example Let $\{X_i\}$ be a sequence of random variables, where $X_i \sim \text{Pois}(\mu = 4)$.
 $E[X_i] = \mu = 4$
 $\text{Var}[X_i] = \mu = 4$
For different values of n (sample size) we will look at the distribution of $Z_n = \bar{X}_n$
(Go to R code)

We have discussed up until now normal approximations for sample averages. We can extend the notion to more general sequences.

Definition 4.6. Asymptotic Normality (AN)

It is said that a sequence of random variables $\{T_n\}$ is asymptotically normal with means $\{\mu_n\}$ and variances $\{\sigma_n^2 > 0\}$ if

$$\frac{T_n - \mu_n}{\sigma_n} \xrightarrow{d} N(0, 1)$$

This denoted as $T_n \sim AN(\mu_n, \sigma_n^2)$, $T_n \sim_A AN(\mu_n, \sigma_n^2)$.

Remark: It is not necessarily true that $E(T_n) = \mu_n$ and that $Var(T_n) = \sigma_n^2$.

The result does say that $\{\mu_n\}, \{\sigma_n^2\}$ adjust $\{T_n\}$ in such a way that $\frac{T_n - \mu_n}{\sigma_n} \xrightarrow{d} N(0, 1)$

Definition 4.6. implies that

$$P(T_n \leq t) \approx \Phi\left(\frac{t - \mu_n}{\sigma_n}\right) \text{ for large } n.$$

In terms of asymptotic normality, the CLT is stated as $Z_n \sim AN(0, 1)$ with $\mu_n \rightarrow 0, \sigma_n^2 \rightarrow 1$ (of definition 4.6)

Using this same AN notation: $\bar{X}_n \sim AN(\mu, \frac{\sigma^2}{n})$ where $\mu_n = \mu, \sigma_n^2 = \frac{\sigma^2}{n}$
Also, $\sum_{i=1}^n X_i \sim AN(n\mu, n\sigma^2)$ where $\mu_n = n\mu, \sigma_n^2 = n\sigma^2$ (All of this using the notation of definition 4.6.)

Theorem 4.6 Slutsky's Theorem.

Suppose that $X_n \xrightarrow{d} X$ ($\lim_{n \rightarrow \infty} G_n(t) = G(t)$ where $G_n(t) = P(X_n \leq t)$, $G(t) = P(X \leq t)$) and $Y_n \xrightarrow{p} c$. Then

1. $X_n + Y_n \xrightarrow{d} X + c$
2. $X_n Y_n \xrightarrow{d} cX$
3. If $c \neq 0$ then $\frac{X_n}{Y_n} \xrightarrow{d} \frac{X}{c}$

Consider a sequence of real numbers b_n such that $\lim_{n \rightarrow \infty} b_n = b$. Trivially, this implies $b_n \xrightarrow{p} b$ (because b_n is not random). Hence, if $X_n \xrightarrow{d} X$, then from Theorem 4.6 $b_n X_n \xrightarrow{d} bX$.

For example: if $X_n \xrightarrow{d} X$ then

$$\frac{n}{n-1} X_n \xrightarrow{d} (1)X = X$$

Example: Asymptotic inference on a population mean.

Let $\{X_i\}$ be i.i.d. with $E[X_i] = \mu_x$ and $Var(X_i) = \sigma_x^2$ both unknown. From the CLT we have that

$$Z_n = \frac{\bar{X}_n - \mu}{\sigma_x / \sqrt{n}} \xrightarrow{d} N(0, 1) (*)$$

$$\widehat{Var(X)} = \tilde{\sigma}_x^2 = \frac{1}{n} \sum_{i=1}^n X_i^2 - (\bar{X}_n)^2 \xrightarrow{p} \sigma_x^2$$

We would like to substitute this estimate in $(*)$ and obtain the same convergence to a standard normal distribution.

In order to do that note that
 $\sqrt{\widehat{Var}(X)} = \sqrt{\tilde{\sigma}_x^2} = \tilde{\sigma}_x \xrightarrow{p} \sigma_x$, i.e. $\tilde{\sigma}_x$ is a consistent estimator of σ_x . (Theorem 4.3.)

Consider $T_n = \frac{\bar{X}_n - \mu}{\tilde{\sigma}_x / \sqrt{n}} = \frac{\bar{X}_n - \mu}{\frac{\tilde{\sigma}_x}{\sqrt{n}}(\frac{\sigma_x}{\sigma_x})} = \frac{Z_n}{W_n}$

Where $W_n = \frac{\tilde{\sigma}_x}{\sigma}$, another application of Theorem 4.3. yields $W_n \xrightarrow{p} 1 \neq 0$, so by Theorem 4.6. (Slutsky's Theorem) (iii) we have that $T_n \xrightarrow{d} \frac{Z}{1} = Z$ where $Z \sim N(0, 1)$

The sequence T_n is called an **asymptotic pivot for μ** because we can use it to draw inferences on μ without having to know population variance. For example, we want 95% C.I. for μ , from the large sample approximation.

$$P(|T_n| \leq 1.96) = Pr\left(\left|\frac{\bar{X}_n - \mu}{\tilde{\sigma}_x/\sqrt{n}}\right| \leq 1.96\right) \approx 0.95$$

after some algebraic manipulations, we obtain:

$$P\left(\bar{X}_n - 1.96 \frac{\tilde{\sigma}_x}{\sqrt{n}} \leq \mu \leq \bar{X}_n + 1.96 \frac{\tilde{\sigma}_x}{\sqrt{n}}\right) \approx 0.95$$

This random interval that will have an approximate probability of 95% of containing the true value of μ .

Now suppose we want to test the hypothesis

$$H_0 : \mu = \mu_0 \text{ vs } H_a : \mu \neq \mu_0 \text{ with a significance level of } \alpha$$

[This is a two-sided test]

The test statistic is

$$T_n = \frac{\bar{X}_n - \mu}{\frac{\hat{\sigma}_x}{\sqrt{n}}} \text{ which under } H_0 \ T_n \sim AN(0, 1)$$

$\therefore$ The asymptotic size α critical region for the test is $R = \{|T_n| > Z_{\alpha/2}\}$
where $P(|N(0, 1)| > Z_{\alpha/2}) = \alpha$ or $P(N(0, 1) \leq Z_{\alpha/2}) = 1 - \alpha/2$